⇒ lowest value (absolute minimum).

Lec # 14:-

Absolute Maximum & Absolute Minimum:-

To find absolute maximum and minimum values of a continuous function f on a closed, bounded set D:

1) Find the value of f at the critical points of f in D.

2) Find the extreme values of f on the boundary of D.

3) The largest of the values from Step 1 and 2 is the absolute maximum value; the smallest of these values is the absolute minimum value.

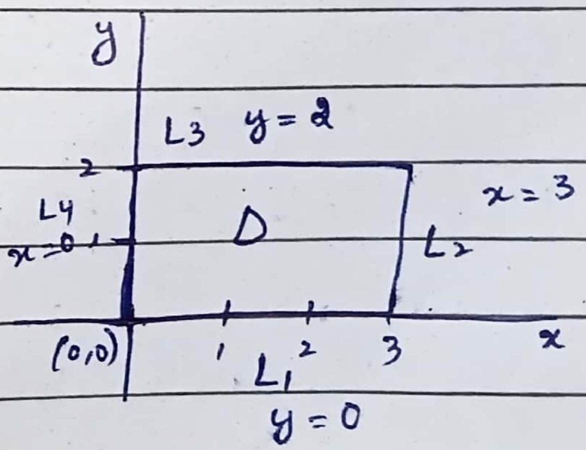
Exp:- Find the absolute maximum and minimum value of the function.

$f(x,y) = x^2 - 2xy + 2y$ on the rectangle.

$D = \{(x,y) \mid 0 \leq x \leq 3 \ , \ 0 \leq y \leq 2\}$

Solution:- Step 1:-

Since f is a polynomial, it is continuous on D.



we first find critical points.

$f_x = 2x - 2y = 0$

$-\cancel{2}y = -\cancel{2}x$

$y = x$

$\boxed{y = 1}$

$f_y = -2x + 2 = 0$

$-2x = -2$

$x = \dfrac{-\cancel{2}}{-\cancel{2}}$

$\boxed{x = 1}$

So, critical point is $(x, y) = (1, 1)$
this point is in D and value of f there
is $f(1,1) = x^2 - 2xy + 2y$

$= (1)^2 - 2(1)(1) + 2(1)$

$= 1 - \cancel{2} + \cancel{2}$

$\boxed{f(1,1) = 1}$

<u>Step 2</u>:- Find the extreme values of f on the boundary of D.

Boundary of D, which consists of four line segments L_1, L_2, L_3, L_4

$\Rightarrow$ on L_1, $y = 0$

$f(x, 0) = x^2 - 2x(0) + 2(0)$

$= x^2 - 0 + 0$

$= x^2$

$0 \leq x \leq 3$

$f(0, 0) = x^2$

$= (0)^2 = 0 \rightarrow$ Minimum

$f(1, 0) = x^2$

$= (1)^2 = 1$

$f(2, 0) = x^2$

$= (2)^2 = 4$

$f(3, 0) = x^2$

$= (3)^2 = 9 \rightarrow$ max

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$\Rightarrow L_x, \quad x = 3$

$$f(3, y) = x^2 - 2xy + 2y$$

$$= (3)^2 - 2(3)y + 2y$$

$$= 9 - 6y + 2y$$

$$= 9 - 4y$$

$$0 \leq y \leq 2$$

$$f(3, 0) = 9 - 4y$$

$$= 9 - 4(0)$$

$$= 9 - 0$$

$$= 9 \longrightarrow \text{Maximum}$$

$$f(3, 1) = 9 - 4y$$

$$= 9 - 4(1)$$

$$= 9 - 4$$

$$= 5$$

$$f(3, 2) = 9 - 4y$$

$$= 9 - 4(2)$$

$$= 9 - 8$$

$$= 1 \longrightarrow \text{minimum}$$

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$\Rightarrow L_3, \quad y = 2$

$f(x,2) = x^2 - 2xy + 2y$

$= x^2 - 2x(2) + 2(2)$

$= x^2 - 4x + 4$

$0 \leq x \leq 3$

$f(0,2) = x^2 - 4x + 4$

$= (0)^2 - 4(0) + 4$

$= 0 - 0 + 4$

$= 4 \quad \rightarrow$ maximum

$f(1,2) = x^2 - 4x + 4$

$= (1)^2 - 4(1) + 4$

$= 1 - 4 + 4$

$= 1$

$f(2,2) = x^2 - 4x + 4$

$= (2)^2 - 4(2) + 4$

$= 4 - 8 + 4$

$= 0 \quad \rightarrow$ Minimum

$f(3,2) = x^2 - 4x + 4$

$= (3)^2 - 4(3) + 4$

$= 9 - 12 + 4$

$= 1$

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$\Rightarrow L_4, \quad x = 0$

$f(0, y) = x^2 - 2xy + 2y$

$= (0)^2 - 2(0)y + 2y$

$= 0 - 0 + 2y$

$= 2y$

$0 \leqslant y \leqslant 2$

$f(0, 0) = 2y$

$= 2(0)$

$= 0 \rightarrow$ minimum

$f(0, 1) = 2y$

$= 2(1)$

$= 2$

$f(0, 2) = 2y$

$= 2(2)$

$= 4 \rightarrow$ maximum.

Step 3:- Find the absolute maximum and absolute minimum at points.

$f(3, 0) = 9 \rightarrow$ Absolute maximum point

$f(0,0) = f(2,2) = 0 \rightarrow$ Absolute minimum point.